

Willow Cunningham

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<https://github.com/willowec>

<https://willowec.net>

Education

University of Maine, Orono

M.S. Computer Engineering, 2024-2025. Cumulative GPA: 4.0

B.S. Computer Engineering, 2020-2024. Cumulative GPA: 3.97

Work Experience

Graduate Research Assistant, UMaine Electrical and Computer Engineering Jun 2024 - May 2025
Orono, Maine

- Helped maintain the Performance Application Programming Interface (PAPI) performance analysis tool.
- Dug into linux kernel code to analyze heterogeneous multicore process scheduling and hardware PMU use.
- Published Works:
 - Willow Cunningham, Vince Weaver. 2024. Performance Measurement on Heterogeneous Processors with PAPI. Presented at the SC '24 Workshops of The Intl. Conf. on High Performance Computing, Network, Storage, and Analysis.
- Technologies: Linux, C, Python

TA / Department Assistant, UMaine Electrical and Computer Engineering Jan 2024 - May 2024
Orono, Maine

- Assisted the supervision and tutoring of an introductory programming lab.
- Designed a battery-powered persistence of vision wand for use at the school's commencement ceremony. The project allowed me to develop CAD skills for use in 3D printing, refined my PCB design skills in KiCad, and provided a variety of software and hardware engineering challenges.
- Technologies: KiCad, FreeCAD, C.

Instrument Software Intern, IDEXX Jun 2023 - Aug 2023
Westbrook, Maine

- Collaborated to develop a full-stack web application monitoring the performance of vet test equipment.
- Designed and implemented an OpenAPI 3.0 compliant API contract.
- Participated in daily SCRUM standup and sprint review.
- Technologies: Python, Flask, Javascript, HTML, Lua, CSS, Docker, SQL, Postman.

Quality Assurance / DevOps Intern, Blue Marble Geographics Dec 2020 - Aug 2022
Brunswick, Maine

- Created a Python wrapper for a C++ GIS SDK along with comprehensive unit tests.
- Generated integration tests for importing/exporting 100+ file formats using the SDK.
- Technologies: Python, C++, Jira, Visual Studio.

Professional References

Professor Vincent Weaver - Advisor | UMaine
vincent.weaver@maine.edu

MacKenzie Meyers - Former Mentor | IDEXX
MacKenzie-Meyers@idexx.com, (603) 998-4362